

Qatar-2b

R-0791

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_240202 · created 2026-07-28T07:33:49+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460342.9539 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.003 +/- 0.066
Transit depth (Rp/Rs)^2	0.0012 +/- 0.046 [%]
Orbital Inclination (inc)	88.31 +/- 1.73
Airmass coefficient 1 (a1)	3.6 +/- 2.34
Airmass coefficient 2 (a2)	-0.55 +/- 0.26
Scatter in the residuals of the lightcurve fit is	62.26 %
Best Comparison Star	#3 - [206, 308]
Optimal Aperture	2.53
Optimal Annulus	5.0
Transit Duration (day)	0.0671 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.003 $\pm$ 0.066 vs 0.16327 (deviation 2.23 σ)
Duration fit vs published	0.0671 d vs 0.0767 d (deviation 2.09 σ)
Mid-time O-C	3.7 min
Depth SNR	0.0
Residual scatter	62.26 %

Light curve

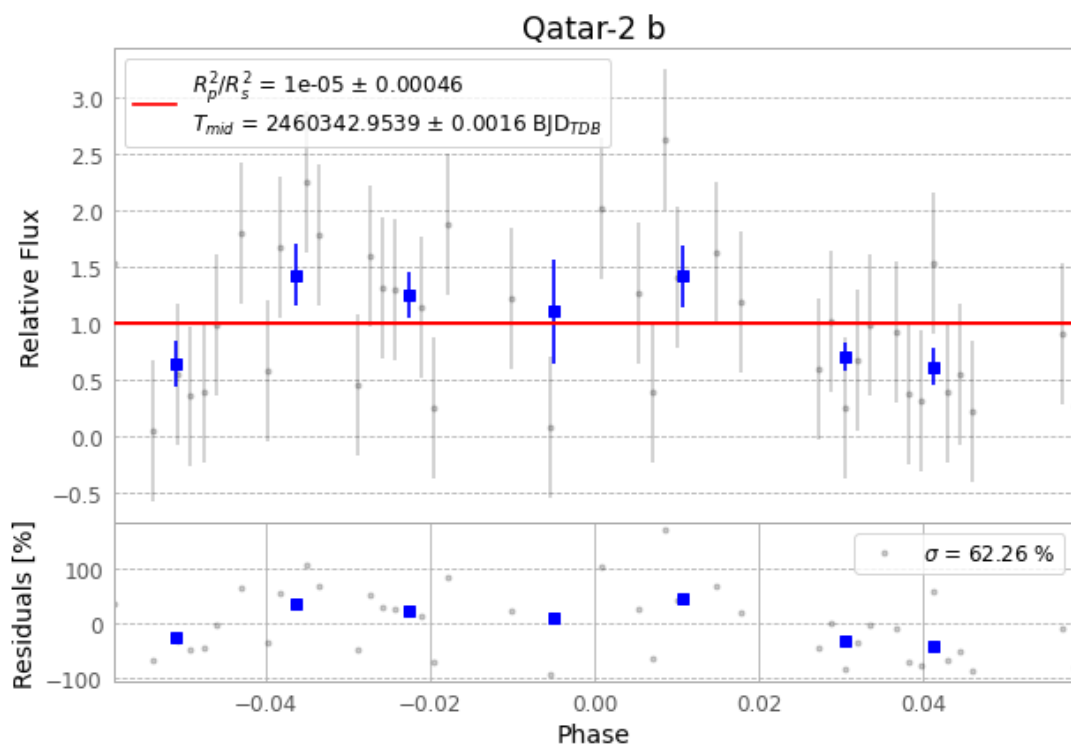


Figure 1 — FinalLightCurve_Qatar-2 b_2024-02-02.png (SHA-256 4d4f60683825b126...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [300, 159], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2836781228020d1c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), Qatar-2240202085416.FITS → Qatar-2240202124215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-240202.log (e28abf149dd4...), FinalLightCurve_Qatar-2 b_2024-02-02.png (4d4f60683825...), FinalLightCurve_Qatar-2 b_2024-02-02.csv (b9b6387382b2...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 07:33Z.